# PRAVHA

**Intelligent Rainwater Harvesting and Adaptive Stormwater Management**

*Every Drop Should Have a Purpose.*

## The Problem

Most rainwater harvesting efforts focus on collecting water, not on managing it before and after storage. This leaves three gaps:

1. **Contaminated first runoff:** the first rooftop flow carries dust, leaves, bird droppings, and other debris into storage.
2. **Unmonitored storage:** there is no automated way to track tank levels or control inflow.
3. **Uncontrolled overflow:** once the tank is full, excess water spills into surroundings, causing waterlogging and adding pressure to drainage systems.

## The Solution

PRAVHA is a low-cost, Arduino-based system that automates the full path of rainwater, from roof to tank to overflow. It uses an **Arduino Uno, one rain sensor, two ultrasonic sensors, and two servo motors.**

### How it works

1. **Detect:** the rain sensor senses rainfall and starts the control process.
2. **Separate:** Servo 1 diverts the initial runoff into a dedicated first-flush chamber.
3. **Filter:** once the first flush is complete, water passes through a filtration chamber.
4. **Store and monitor:** filtered water enters the main tank, where Ultrasonic Sensor 1 tracks the water level.
5. **Divert:** when the tank reaches its maximum operating level, Servo 2 redirects incoming water away from it.
6. **Manage overflow:** Ultrasonic Sensor 2 monitors the overflow or diversion path.
7. **Communicate:** an LCD and LED indicators show system status, and push buttons allow reset and manual override.

## Where It Can Be Used

- **Rural areas:** excess water is routed to groundwater infiltration structures such as recharge pits or trenches.
- **Urban areas:** excess water is routed to rain gardens or sponge gardens, which help absorb runoff and ease pressure on drains.

## What Makes It Different

PRAVHA does not claim to invent new components. Its originality lies in **system-level integration**: first-flush separation, filtration, automated level monitoring, and controlled overflow diversion combined in one adaptable architecture that can be configured for different settings.

## Objectives

- Improve the quality and utilization of harvested rainwater
- Reduce unmanaged runoff and localized waterlogging
- Support sustainable, decentralized water management

*These benefits are expected outcomes and will require experimental validation through prototype testing.*

**PRAVHA: rainwater should not simply be collected. It should be managed intelligently.**